Santiago Páez Avendaño

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EDUCATION

PhD Student in Astrophysics Aug. 2026 – Present
Instituto de Astronomía, Universidad Nacional Autónoma de México (UNAM), Mexico City, México
Advisor: Yilen Gómez Maqueo Chew

M.Sc. in Astrophysics *(with honors)* Aug. 2024 – Aug. 2026
Instituto de Astronomía, Universidad Nacional Autónoma de México (UNAM), Mexico City, México
Thesis: Data reduction method for OPTICAM multiband time series of transiting exoplanets.
Advisor: Yilen Gómez Maqueo Chew. GPA: 9.44/10.0.

B.A. in Natural Sciences Education Jan. 2019 – Mar. 2024
Universidad de La Sabana, Chía, Colombia
Awarded thesis: Exoplanets Hackathon: a challenge-based learning proposal to develop scientific skills.
Advisor: David Tovar Rodríguez. GPA: 4.16/5.0.

RESEARCH AND ACADEMIC EXPERIENCE

PUBLICATIONS AND WORK IN PREPARATION

- **Páez**, **S.**, Gómez Maqueo Chew, Y., and Hebb, L. (2026). "Data reduction method for OPTICAM multiband time series of transiting exoplanets". *RAS Techniques and Instruments* 5, rzag021.
- Khandelwal, A. et al. [included **Páez, S.**] (2026). "TOI-7413 b: A super-Neptune in the Neptunian ridge orbiting and M dwarf". *in prep.*
- Khandelwal, A., Das, S. N., et al. [included **Páez, S.**] (2026). "TOI-6884b: a low-mass brown dwarf transiting a slightly evolved star". *Monthly Notices of the Royal Astronomical Society* 549(4), stag988.
- Gómez Maqueo Chew, Y. et al. [included **Páez, S.**] (2026). "TOI-1080 b: a temperate, rocky planet orbiting a quiet M4V host". *Monthly Notices of the Royal Astronomical Society* 548.

OBSERVATIONAL EXPERIENCE

- **SAINT-EX project.**
  Observer, SAINT-EX/OAN-SPM 1 m Telescope. Time: 19 nights between Aug. 2026 and Oct. 2026. Role: Launching remote observations.
- **Insights into stellar activity cycles: Spot crossing during transit.**
  Co-I, OPTICAM/OAN-SPM 2.1m Telescope. Time: 3 nights in Mar. 2025 and 3 nights in Mar. 2026. Role: Data acquisition and data reduction.
- **Mass Measurements of Two Gas Giant Planets around M-dwarf stars.**
  Co-I, WIYN/NEID. Time: 1.75 nights in 2026A. Role: Preparation of observing proposal and transit fitting.
- **Multi-band Transit Follow-up of Giant Planet Candidates Around M-Dwarfs with OPTICAM.**
  Co-I, OPTICAM/OAN-SPM 2.1m Telescope. Time: 6 nights in Feb. 2025 and 6 nights in Feb. 2026. Role: Preparation of observing proposal, design of observation plan, data acquisition, data reduction, and transit fitting.

SELECTED PRESENTATIONS

- Ground-based multiband transit follow-up with OPTICAM and SAINT-EX at OAN-SPM. *Poster at 2026 Sagan Exoplanet Summer Workshop*. 2026.

- Validation of TOI-7413 b: A Neptunian Planet on the Neptunian Ridge Around an M Dwarf. *Poster at 2026 Sagan Exoplanet Summer Workshop*. 2026.
- Data reduction method for OPTICAM multiband time-series of transiting exoplanets. *Oral presentation in the XXII Astronomy Students Meeting (REA)*. 2025.
- Método de reducción de datos para series de tiempo multibanda con OPTICAM de exoplanetas transitantes. *Seminar of Astrophysics and Cosmology COSMOS - Instituto de Ciencias Físicas - UNAM*. 2025.
- Proyéctate: a novel strategy to reach the public audience using Astronomy as a catching science topic. Oral Presentation. *Communicating Astronomy with the Public - CAP*. 2024.
- Follow-up of WASP-43b and WASP-107b transits through differential photometry from OAN-TNT. Poster. *LXVI National Physics Congress*. 2023.

SELECTED WORKSHOPS AND SCHOOLS

Code/Astro Workshop Aug. 4–8, 2025
One week of training in astronomy software development: testing, documentation, profiling, parallelization, and open-source packaging.

XV ESAOBELA Jan. 2023 – Feb. 2023
100 hours of observation with 1-m and 10-inch telescopes and 51 hours of theoretical instruction at OAN-Tonantzintla.

XXXI Scientific Summer at the OAN-SPM Jun. 2022
Astrophysics training and remote observation with the OAN-SPM 84-cm telescope.

Alpha-Cen Summer School Aug. 2021
Programming for astronomical data analysis and an introduction to core astrophysics topics.

RECOGNITIONS AND SCHOLARSHIPS

National Graduate Scholarship (SECIHTI), master's degree Aug. 2024 – Jun. 2026
Global Student Prize, Top 50 finalist Jul. 2023
Thesis IN-BITE, second place, inter-university thesis competition Nov. 2023
Best Saber Pro score, Natural Sciences Education Oct. 2023
FEST Scholarship, fully funded undergraduate scholarship Jan. 2019 – Nov. 2024

OUTREACH

- Organized and participated in eight astronomy festivals in Sabana Centro, Colombia: over 4,000 attendees (2019–2023).
- Public talks on exoplanets: Clavius Astronomical Center, Universidad Iberoamericana (2026), and Orizaba Planetarium (2025).
- Outreach staff at the Guillermo Haro Mobile Planetarium during the Noche de las Estrellas festival (2025).

SKILLS AND LANGUAGES

Programming and tools: Python, Git/GitHub, LaTeX, Linux, macOS.
Software: Jupyter Notebook, AstroImageJ, TOPCAT, Poetry, UV.
Scientific packages: Astropy, photutils, SciPy, juliet, exoplanet, pyaneti, PyMC.
Languages: Native Spanish; English B2.